FRANCK PORTIER
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M1 MACRO
Group B
Session 5

These notes are written during class. They are posted "for the record" and may content mistakes. They are not intented to be read by anyone that did not attend the class.

Government

$$\begin{cases} G_1 = T_1 + B \\ G_2 + (1+r)B = T_2 \end{cases}$$

B = public debt

$$\underbrace{G_1 - T_1}_{\text{Budget deficit}} \geq 0 \Rightarrow B \geq 0$$

IBC

$$\underline{G_1} + \frac{\underline{G_2}}{1+r} = \underline{T_1} + \frac{\underline{T_2}}{1+r} \qquad \underline{\ \ } = \text{given}$$

Households

$$\begin{cases} U = \log c_1 + \beta \log c_2 \\ c_1 + \dfrac{c_2}{1+r} = \underbrace{(Y_1 - T_1)}_{\text{disposable income of period } \underline{1}} + \dfrac{(Y_2 - T_2)}{1+r} \end{cases}$$

FOC for HH ?

$$u = \log c_1 + \beta \log\left[(1+r)(y_1 - T_1) + y_2 - T_2 - (1+r)c_1\right]$$

FOC : $$\frac{1}{c_1} = \beta(1+r)\frac{1}{c_2} \quad \text{(as before)}$$

$\hookrightarrow$ Euler intertemporal equation

Eq conditions

good market in 1 : $y_1 = c_1 + G_1$

2 : $y_2 = c_2 + G_2$

Solving for the equilibrium

$$C_1 = Y_1 - G_1$$
$$C_2 = Y_2 - G_2$$

[unknowns: C_1 C_2
$1+r$
(or $\frac{P_2}{P_1}$)]

Therefore, pluging C_1 and C_2 in the FOC:

$$\frac{1}{C_1} = \beta(1+r)\frac{1}{C_2}$$

$$\frac{1}{Y_1-G_1} = \beta(1+r)\frac{1}{Y_2-G_2} \Rightarrow \boxed{r = \frac{1}{\beta}\frac{Y_2-G_2}{Y_1-G_1} - 1}$$

$$\begin{cases} r = \dfrac{1}{\beta}\dfrac{Y_2 - G_2}{Y_1 - G_1} - 1 \\ C_1 = Y_1 - G_1 \\ C_2 = Y_2 - G_2 \end{cases} \Rightarrow \text{solution}$$

$$\underbrace{G_1 + \frac{G_2}{1+r}}_{\text{Disc. sum of spendings}} = \underbrace{T_1 + \frac{T_2}{1+r}}_{\text{Disc. sum of taxes}}$$

$$G_1 + \frac{G_2}{1+r} = \tilde{T}_1 + \frac{\tilde{T}_2}{1+r}$$

$$\tilde{T}_1 < T_1$$

$$\tilde{T}_2 > T_2$$

RICARDIAN EQUIVALENCE (BARRO)